Chapter 12

FISH CULTURING

Fish Culturing

Fish culture, also known as fish farming or aquaculture, is the practice of raising fish in controlled environments for commercial, scientific, or recreational purposes. It involves the cultivation of fish, crustaceans, molluscs, and other aquatic organisms in tanks, ponds, or other aquatic habitats, under controlled conditions. Fish culture can be done in both freshwater and saltwater environments and can involve various species of fish, including tilapia, trout, salmon, catfish, and carp. The practice of fish culture plays an important role in global food security by providing a source of protein for human consumption and also contributes to the conservation of wild fish populations by reducing pressure on wild stocks.

In Country like Pakistan fish has special importance as a supplement to ill balanced cereal diets. Today protein deficiency is the world's most serious human malnutrition problem, and perhaps 30 to 40% of the world population is suffering from protein deficiency. It is estimated that about 10 million tons of fish is required annually to meet the present day demand of fish protein in the country against an annual production of only 3.5 million tons. Many of the water reservoirs remain either unused or not properly used for fish culture for want of proper scientific know how. In recent years researches conducted by the Fisheries Research Institute have revolutionized fish culture in Pakistan.

Evolution of Fish Culture

From ancient times man has utilized the ponds and kept fishes for their interest. Afterwards they started to use the fishes for food. It was a problem before them as how to increase the fishes in number and maintain them in ponds. Previously nothing was known about the environmental and physiological conditions of fishes hence the process of fish culture was unsuccessful. Thus the workers in their anxiety to know more about the fishes started to study the morphology, physiology, histology, habit and habitat of the fishes, and concluded that

(1) Fishes are growing themselves.

(2) Fishes are breeding themselves.

These findings proved to be keynote and encouraged them to think about the fish culture at commercial level. Now-a-days fish culture is expanding under the pressure of growing population in China, Indonesia, Vietnam, Africa, and Cambodia, U.S.A. and Pakistan. The cultivation of a number of economically useful fish species, artificial fertilization, induced breeding etc. has opened a new era in the fish culture programme.

Aim of Fish Culture

(1) The main aim of fish culture is to obtain maximum yield of fish.
(2) To obtain palatable and highly nutritive fish flesh.
(3) By products of fishing industry.

Qualities of Culturable Fishes

The main attraction of fish culture is to obtain more fish. So, before starting this complicated programme it is essential to have a comprehensive idea about the fish itself. The culturable fishes,

(1) Should have ability to feed on natural food.
(2) Should have ability to feed on artificial diet.
(3) Should consume small quantity of food for development.
(4) Should be herbivorous in nature.
(5) Should be able to tolerate a sudden change in climatic conditions and nature of water in ponds.
(6) Should be fast growing and can attain good length and weight both.
(7) Should live in ponds with other, fishes without any disturbance.
(8) Should be able to resist against diseases.
(9) Should be prolific breeder.
(10) Should be palatable and much nutritive.

Types of Cultivable Fishes

The cultivable fishes are of 3 types:

(1) Indigenous or native fresh water fishes viz., major carps.
(2) Salt water fishes acclimatized for fresh water viz., Chanos, Mullets.
(3) Exotic fishes, imported from other countries viz., Mirror carp, Chinese carp, Crucian carp and Common carps.

Types of Culture in Pakistan

It is practically difficult to find all the characteristics of a culturable fish in any one of the fishes but some fishes are found most suitable for fish culture.

Among these, 'Major Carps' have proved to be best culturable fish in Pakistan having following qualities.

(1) Carps feed on zoo-and phyto-planktons, decaying weeds, debris and other aquatic plants.
(2) Carps can survive under higher temperatures and also in turbid water.
(3) Carps can tolerate oxygen variation in water.
(4) Carps have fast growth rate and can attain good size and weight.
(5) Carps can be transported from one place to other easily.
(6) The flesh of carps is mostly palatable and much nutritive.

External Factors Affecting Fish Culture

Now-a-days, profitable fish culture has become purely technical and scientific because a number of environmental factors affect the fish culture programme. A number of factors viz., temperature, light, rain, water, flood, water current, turbidity of water, diseases, toxic pollutants, pH, hardness and salinity of water, dissolved oxygen etc. play a vital role in fish culture. The temperature and light intensity both have profound effect because fish cannot breed above or below the critical temperature which may be variable according to the species of fish.

◉ Management of Fish Culture Programme

Fish culture is a complicated process, so for an ideal fish culture one should have an idea about the different stages of fish culture i.e., topographic situation, quality of water, source of water, and other physical, chemical and biological factors. Ponds are the sites where fish develops and grows. The management of ponds has its own importance and has to be tackled from the point of view of breeding, hatching, nursing, rearing and stocking ponds. The nature of ponds may slightly vary with the species on the fish. Even different stages of the same fish are cultured in the ponds having quite different properties. Keeping in view the various stages of fishes, the following different types of ponds have been recommended to manage them.

◉ Breeding Pond

First step in the fish culture is the breeding of fishes, therefore, for proper breeding special types of ponds are prepared called as breeding ponds. These ponds are prepared near the rivers or other natural water resources.

◉ Types of breeding

According to the mode of breeding there are two categories:

I. Natural breeding (Bundh breeding)

The natural bundhs are special types of ponds where natural riverine conditions or any natural water rqsource conditions are managed for the breeding of culturable fishes. These specially designed bundhs are constructed in large low-lying area having facility to accommodate large quantity of rain water. These bundhs are having an outlet for the exit of excess rain water. The shallow area of such bundhs is always used as spawning ground. These bundhs are of three types:

(1) Wet bundh.

(2) Dry bundh.

(3) Modern bundh.

1. **Wet bundh.** The ponds specially constructed for fish breeding having water throughout the year are known as wet bundhs or perennial bundhs. The flow of water from outlet is controlled with the help of bamboo fencing.

Dry bundh. This type of ponds is purely seasonal with shallow water areas. This is constructed by keeping soil walls from three sides and open area from one side. In monsoon period rain water flows towards this bundh and fills the pond. But after monsoon water this bundh dries up after a month or two.

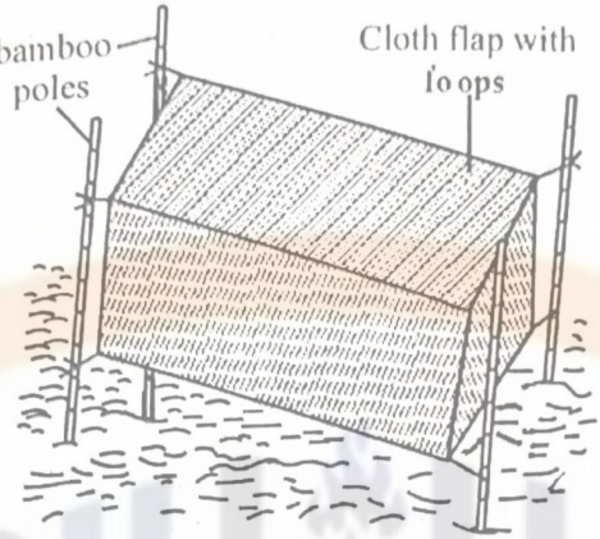

Fig 12.1: Wet bundh

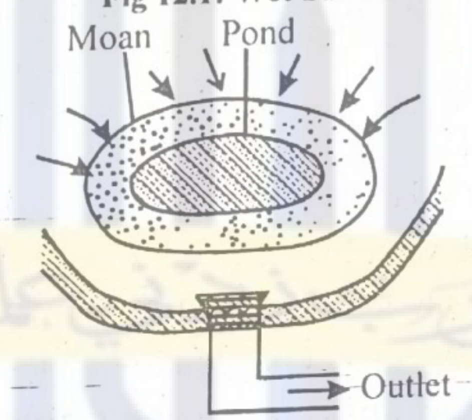

Fig 12.2: Dry bundh

2. **Modern bundh.** This is known as 'Piicca bundh'. It is a masony construction and a sluice gate at the lower-most level of the bundh is the characteristic feature. The total exit of water from the bundh is possible by this gate so that after each spawning, bundh is cleared of water from the bundh. An inlet is formed at the higher level of bundh for the entrance of the water while an outlet is prepared in low lying area for the exit of the water from the bundh.

II. Induced breeding

The fish seed is commonly collected from breeding grounds but this has certain drawbacks because a mixture of eggs of various types of fishes is obtained in which some eggs may be of predatory fishes. Keeping this in mind some advanced techniques have been devised for the super quality of fish seed by artificial method of fertilization.

Artificial fertilization: In this method males are taken out by artificial way and the eggs are got fertilized by the sperms. This is of two types:

Dry method. Eggs and sperms of fishes (which are just taken out) are mixed thoroughly and left in this condition for about 30 minutes. Now water is added in this mixture after the fertilization of eggs.

Wet method. Eggs are kept in water and milt of the male is spread directly into this water. This method is commonly used in sticky eggs.

For artificial fertilization, mature female fish having eggs is taken in hand and dried by towel. The belly of this fish should be upward. Stripping should be started with the thumb of the right hand from the anterior to the posterior direction for the ejection of the eggs due to force. In this way eggs are collected separately. Further, a male fish is caught and its belly is kept downward. The milt of the fish is stripped with the help of the thumb and index finger and collected separately. Now the eggs are subjected to fertilization as mentioned above.

Types of induced breeding. In accordance with the breeding habit, two types of fishes are found **i.e.**, fishes commonly breeding in ponds and fishes not commonly breeding in ponds. Therefore, induced breeding is managed in both types of fishes by different methods as described below. Method of inducing in pond breeding fishes. Although pond breeding fishes are well adapted for breeding in the maintained ponds yet for effective and successful breeding some inducement is given to the pond breeding fishes also. First of all they are segregated sex-wise to prevent wild spawning. Further, they are allowed to breed under the influence of any one of the following spawning stimulants:

(i) By the introduction of specific selected brood fishes in breeding ponds,

(ii) By replacement of old water by fresh rain water,

(iii) By providing suitable site for the attachment of the laid eggs,

(iv) By providing particular temperature and light intensity.

1. **Method of inducing in difficult spawning fishes.** The fishes like Chinese carp, Cat fishes, Mullets, Milk fish etc. do not commonly breed in ponds, therefore, some breeding stimuli are needed for proper breeding.

(a) Induced bundh breeding. Spawning in bundhs occurs after continuous heavy showers when rain water covers the bundh. The breeding of carp is very much interesting. The male and the female indulge in sex play. The male starts chasing the female with vigorous splashing of water. The female is then held by the male and ova are released, at the same time male ejects the milt over the eggs. After spawning the breeders remain guilt for some time and then swim away. The bundh breeding of 'hardly spawning' is induced by the following factors :

(1) Spawning is stimulated by the following monsoon.

(2) Water current also induces spawning.

(3) Temperature between 24 to 32°C is found best inducer for spawning.

(4) Cloudy season followed by thunderstorm and rain also stimulates spawning.

(b) Induced breeding by hypophysation (Hormone). The gonadotropin hormone (F.S.H. and LH) secreted by pituitary gland influences the maturation of gonads and spawning in the fishes.

Method of hypophysation. First of all pituitary is taken out, then preserved in absolute alcohol inside the sealed tube in a dessicator at room temperature or in acetone for about 36 hours and stored in sealed phails in refrigerator. The pituitary glands should be taken out form fully mature, healthy and freshly killed fishes. The donor fish of the same species are most preferred for this purpose.

For the preparation of extract, the weighed glands are homogenized in distilled water of 0.3% saline or glycerin and centrifuged for 15 minutes at 20,000 rpm. The supernatant thus obtained is injected intramuscularly on the back or on the base of the caudal fin. In some cases intraperitoneal injection is also recommended at the base of the pectoral fin.

Dose for hypophysation. For the first time 2 to 3 mg of pituitary extracts/kg body weight of fish is injected. After 6 hours of first injection, slightly higher dose of 5 to 8 mg/kg body weight of female and 2 to 3 mg/kg body weight of male are injected. Now both male and female are released into a 'cloth hapa' which is fixed with bamboo poles in the breeding pond. The breeding hapa is a rectangular box formed of mosquito net cloth and closed from all sides.

After 24 hours of injection the male and the female start to swim, become excited and restless and always try to jump out of the hapa. The male and female chase each-other and push each-other with their snouts and spawning occurs as result of which eggs are fertilized. Now the fertilized eggs are removed from spawning place and kept into hatching hapas.

Synthetic hormone. With the advancement of fish culture programme the synthetic hormones have been introduced in the field for breeding of fishes. The human chorionic gonadotropin (HCG) and some cortic steroids particularly 'Deoxy corticosterone are most successfully used for induced spawning of cultivable fishes. A mixture of chorionic gonadotrophic and mammalian pituitary extract as 'Synaphorin' is used in combination with the fish pituitary extract.

Fish Seed

The fish seed is collected from the breeding ponds. The best suited places for the collection of fish seeds are on the curve area of these rivers. The spawn collecting net is commonly called as 'Benchi Jal (Shooting net). The shooting net is a funnel shaped structure consisting of closed mesh or cheap coarse cloth (Fig. 12.3).

The whole net is 10 to 20' 'in length and both the ends i.e., anterior and posterior are open. At the posterior end of the net a tailed-structure is attached which is a rectangular piece (2'xl'xT) called, as 'Gamcha'. All stages of fish gamcha cart be collected in this gamcha from where they are scooped and transferred to the 'Hatching pits. Hatching Pit

The fertilized eggs are kept into hatching pits for hatching. At the time of construction of hatching pits one should see that the hatching pit:

(1) Should be nearer to the breeding grounds.
(2) Should be smaller in size.
(3) Should contain such quantity of water which must dry within 1 or 2 month.
(4) Should be more in number.

Fig 12.3: Hatcharies

Types of hatching pits

Hatching pits are of two types:

1. **Hatcharies.** These are small sized ponds in which fertilized eggs are transferred. After 2 to 15 hours the fertilized eggs are hatched. Some draw backs make the hatcheries unfit for advanced fish culture programme. These drawbacks are:

   (i) Sudden rise and fall in temperature.

   (ii) entrance of predators in ponds.

   (iii) Drying of water from ponds may cause mortality of eggs.

To overcome all these drawbacks specially designed hatcheries and hatching hapas are constructed.

Hatching hapas. Hapas are rectangular trough shaped tanks made up of cloth supported supported by bamboo poles fixed in the river. In these hapas fish eggs are aerated by continuous flow of current. The size of hapa is about 3" X 1.5"x 1' and is made up of mosquito net cloth which is fixed into outer larger hapa made up of coarse cloth. Two types of hapas are designed:

(a) ***Fixed type hapa.*** If possible to fix the perpendicular poles (arms of hapa) only then the fixed type of hapa is used for hatchings of eggs otherwise floating type of hapas should be used.

(b) ***Floating type hapa.*** In hard bottom areas where it is not possible to use fixed hapas, floating hapas are built. These hapas are arranged in large number in series attached with the bamboo and floated on water surface. To avoid overcrowding of eggs. only one layer of eggs is spread in each hapa. Hatching of eggs takes place in the outer hapas leaving the egg membrane in the inner one. The hatchings are kept for 36-48 hrs in hapas then transferred to nurseries.

Fig 12.4: Hatching hapas

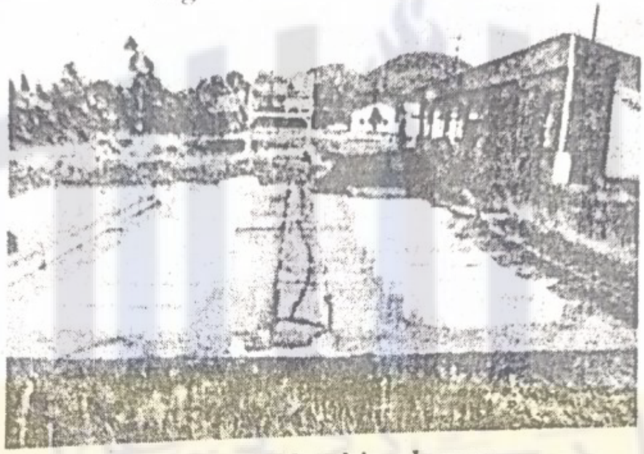

Fig 12.5: Hatching hapas

Transport of Fish Fry to Nursery Ponds

The fish fries are collected and transported to nurseries. These are transported in earthen 'Hundies' which cause heavy mortality due to following factors.

(1) Decreased dissolved oxygen concentration in water.

(2) Increased carbon dioxide concentration in water.

(3) Toxicity of wastes like excreted ammonia.

(4) Hyperactivity and its strain.

(5) Physical injury to fries during transport.

To overcome the above mentioned injuries, fries are subjected to conditioning before transportation by keeping them into fixed volume of water for definite period and then transported in open or closed vessels. Temperature of the water of the vessels can be kept lower by tieing a wet cloth around the vessel. Dead fries should be removed to avoid the pollution and infections. Now-a-days fries are transported in sealed metal containers with oxygen. In India alkethen bags of different sizes are used for the transportation of fish fries from hatching hapas to nursery ponds.

Nursery Pond

The newly hatched fries transported from hatching hapa to nursery ponds are very tender so one should be very careful in maintaining them there. Nursery ponds should always be near the hatching hapas. The nursery ponds are small small set of shallow (3' to 5' in depth) water reservoir. But now-a-days generally more deep (5' to 8') ponds are prepared having an area of about 1/2 acre. The ideal and recommended nursery ponds are 50 to 60' x 30 to 40' x 4 to 5'. The nursery ponds should be prepared before the hatching. The exit and entrance of water should also be under control. In nursery ponds natural resources of food are less and when large numbers of fries are released they certainly suffer due to lack of food. First of all predatory and weed fishes should be removed from pond. The chemical fertilizers ammonium sulphate, sodium nitrate and superphosphate should be used along with cow dung. The amount of chemical fertilizers should be less in quantity: 5,000 to 70,000 kg cow dung/hectare is recommended in accordance with nature and fertility of the pond soil. Due to these manures numerous zooplanktons are developed within 10 to 20 days and phytoplankton are also grown up within10 to 15 days. On these phyto and zooplanktons they can feed easily comfortably.

Mortality in nursery pond

The heavy mortality of fish fries has been recorded in nursery ponds. The following factors are responsible for such mortality:

(1) Sudden change in the quality of water form hatching hapa to nursery pond
(2) Lack of suitable food in pond.
(3) Presence of predatory fishes and predatory aquatic insects in the pond.
(4) Overgrowth of plankton. Decreased oxygen concentration in water.
(5) Cannibalism.

Fig 12.6: Nursery pond

Precautions for nursery ponds

(1) In the nursery ponds water should be under good control and circulating.
(2) Pond should be nearer to the hatching ponds.
(3) Ponds should be predator free.
(4) To avoid the overcrowding, fries should be kept in limited number.
(5) Supply of food material should be proper.
(6) When fries are more developed in the nursery ponds, and attain a length of 10 to 15 cm they should be transferred into the rearing ponds.

◉ Types of Ponds

There are four types of fish ponds: Barrage, excavated, elevated and combination excavated/elevated.

1. **Barrage Ponds**: These types of ponds are most appropriate in hilly areas. This type of pond is constructed by building a dam across a low point in a valley that may have intermittent, water flow. The barrage dam captures the surface runoff, and a crop of fish can be raised in this water. This type of pond is the least expensive to construct and most appropriate for hilly area, however, it is more difficult to manage than ponds with a more regular shape.

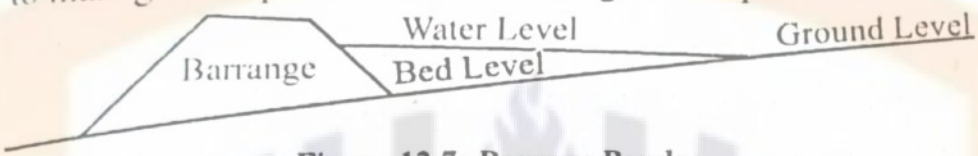

Figure 12.7: Barrage Ponds

2. **Excavated Ponds**: these are most appropriate in low lying areas. Often this pond type will not be drainable because of natural seepage into the pond, because of the inability to dry the pond bottom; these low-lying sites are generally least preferred to build the fish farm.

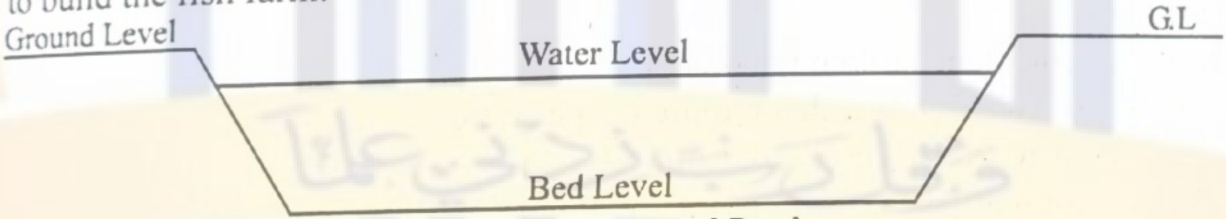

Figure 12.8: Excavated Ponds

3. **Elevated Ponds**: Can only be built when the soil contains high clay content. The water supply almost always has to be pumped; however, gravity can often be utilized for drainage.

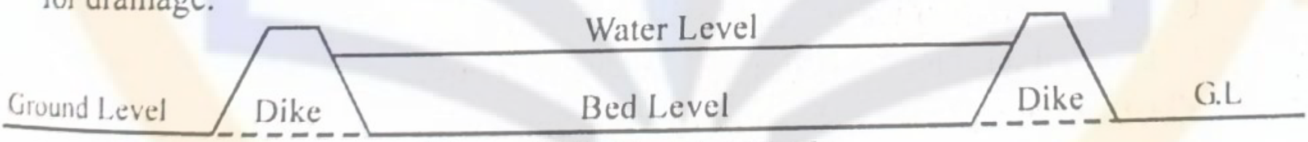

Figure 12.9: Elevated Ponds

4. **Excavated / elevated Ponds**: the most appropriate pond type for most areas of the Punjab is the combination excavated / elevated pond. If the soil has sufficient clay content, the dikes can be built from the soil that is removed during pond excavation, thus excavation costs are minimized. Additionally, because the pond is not completely excavated.

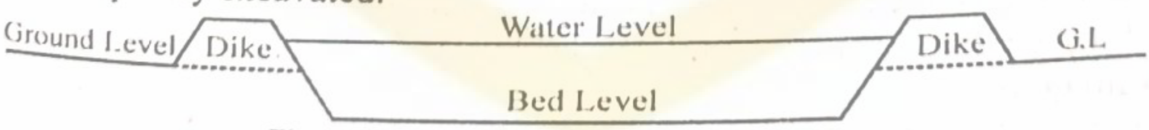

Figure 12.10: Excavated / elevated Ponds

Ponds can also be divided on the basis of constructed materials. Following are the types of ponds according to their material types.

◉ **Earthen ponds:** These are the most common type of fish ponds and are typically made by excavating a depression in the ground and lining it with clay or a synthetic liner to prevent water from seeping into the soil. They are usually shallow and can be used to raise a variety of fish species. Earthen ponds are relatively inexpensive to construct and maintain and can be easily integrated into the landscape. However, they are vulnerable to erosion, leaks, and soil contamination.

Advantages:

- Relatively inexpensive to construct and maintain
- Can be easily integrated into the landscape
- Can support a wide range of fish species
- Can provide natural food sources for fish such as plankton and insects

Disadvantages:

- Vulnerable to erosion, leaks, and soil contamination
- Water quality can be difficult to control
- May require periodic dredging to maintain depth
- Not suitable for fish that require deeper water

◉ **Concrete ponds:** These are made of concrete and are more durable than earthen ponds. They can be used for raising fish that require deeper water, such as catfish. Concrete ponds are more expensive to construct than earthen ponds but are more durable and can last for many years. They are also less vulnerable to leaks and soil contamination

Advantages:

- More durable than earthen ponds
- Can be used to raise fish that require deeper water
- Water quality can be easily controlled
- Can last for many years with proper maintenance

Disadvantages:

- More expensive to construct and maintain than earthen ponds
- Can be difficult to integrate into the landscape
- Can be prone to cracking and leaks
- Can be vulnerable to extreme temperature changes

◉ **Plastic-lined ponds:** These ponds are similar to earthen ponds, but are lined with a plastic liner instead of clay. They are easy to construct and maintain, and can be used for raising a variety of fish specie

Advantages:

- Easy to construct and maintain
- Can be used to raise a variety of fish species
- Less vulnerable to erosion and leaks than earthen ponds
- Can be relocated if necessary

Disadvantages:

- Vulnerable to punctures and tears in the liner
- May require replacement of the liner periodically
- May not be suitable for fish that require deeper water
- May be less aesthetically pleasing than other types of ponds

◉ **Raceways:** These are long, narrow channels that are typically made of concrete or other materials. Water is circulated through the raceway, and fish are raised in the flowing water. They are commonly used for raising trout and other species that require high oxygen levels. Raceways can be more expensive to construct and maintain than other types of ponds, but they offer excellent water quality and efficient use of space.

Advantages:

- Excellent water quality and oxygenation
- Efficient use of space
- Can be used to raise fish that require high oxygen levels
- Can be used in a recirculating aquaculture system

Disadvantages:

- More expensive to construct and maintain than other types of ponds
- Can be difficult to clean
- May require constant monitoring to maintain water quality
- Not suitable for fish that require still water.

◉ **Cage ponds:** These are floating structures that contain netted enclosures where fish are raised. They are commonly used for raising salmon and other marine species in open water. Cage ponds are more expensive to construct and maintain than other types of ponds, but they offer excellent water quality, efficient use of space, and protection from predators.

Advantages:

- More durable than earthen ponds
- Can be used to raise fish that require deeper water
- Water quality can be easily controlled
- Can last for many years with proper maintenance
- with proper maintenance

◉ Rearing Ponds

For good health and growth of fingerlings the exercise is essential for them inside the rearing ponds. So these fingerlings are reared in longer and narrower ponds to provide them long distance for swimming. The water of this pond may be seasonal or perennial. The rearing ponds should be free from toxicant and predators! The depth of the pond should be about six feet containing nutritive food material in accordance with the population of fingerlings. As fingerlings attain a length of about 20 cm they should be transferred to the other type of ponds called as stocking ponds.

Fig 12.7: Rearing pond

Transport of fingerlings

The fingerlings are transported form rearing ponds to stocking ponds in a container of 1,000 liter capacity. This container is internally lined with foam to avoid physical injury. The proper arrangement for aeration of the tank is essential during transport. To make the fingerlings inactive various sedative (sodium amylate and barbifurate) are used. This is only for less consumption of dissolved oxygen during transport. Sometimes some diseases, parasites and predators are also transported along with fingerlings. To avoid these fingerlings should be washed carefully before packing and the use of antibiotics, methyl blue, copper sulphate, potassium permanganate, formalin and common salt are recommended. Taking all these precautions, the fingerlings are transferred to the stocking ponds.

Stocking Ponds

The stocking ponds should be cleaned of weeds and predatory fishes.

The sufficient food is essential for good growth of fishes in these ponds hence proper manuring should be done to increase the production of zooplankton o nursery ponds because large number of fishes may not be able to feed properly due to lack of food material. As for the proper organic manuring cow dung is the best; in the pond and should be used at the rate of 20,000 to 25,000 kg/hectare/year. The inorganic chemical fertilizers are also used super-phosphate, ammonium nitrate and ammonium sulphate at the rate of 1,000 to 1,500 kg/hectare/year. The powdered rice, paddy, oil cakes, coconut, mustard, groundnut, etc. are commonly used as artificial food for the fishes. The artificial food used for the fishes should be easily digestible in natural form and economically suitable, the best time for feeding the fishes is in the morning hours. The quality of food should not be changed suddenly. The amount of fertilizers used is totally dependent on the fertility of the soil, number of fishes and types of fishes being kept in the stocking ponds. When fishes attain maximum length and weight they should be harvested.

Fig 12.8: Stocking pond

Harvesting

Harvesting is done to capture the fishes from the water. The well grown fishes are taken out for marketing and smaller ones are again released into stocking ponds for their growth. In highly organized and well planned fish farming the fishes below a particular size are not generally captured.

Methods of Fishing

For proper and scientific management of fishing from different resources, the idea of fishing gear and net in accordance with the size and habitat of fish is essential. The oldest art of fishing in the form of stones and spears has been converted into complex traps, nets and lines. In our country, though a large number of fishing methods are in use but all are not of common use throughout the country. Some common methods of Fishing are described here:

Stranding. It is widely used in shallow water resources in which some of the shallow area of the pond is separated from the main body by throwing up a low earthen bundh when almost all the water is thrown out from this area, fishes are gathered and caught out. This method is economical but can only be used in shallow swamps and burrow pits.

Angling. This is also very common in practice for fishing of large and predatory fishes. Along the banks of rivers of ponds fishing rods are left dug into the soil and the owner of rod visits to unhook the occasional catch or renew the bait consisting of earthworms. Sometimes very long lines containing 100 to 250 hooks attached to it, are also used for fishing. This method can be used both in shallow and deep waters.

Traps. This may also be used in shallow and deep waters both. Traps are made of basket work of varying materials. They vary in shape, size and mode of operation. The cage trap is of fixed type whereas scoop and cover traps are movable at the time of operation. Cover traps are under common practice throughout the Country and specially used in muddy bottom. The basket is plugged into the water where fish is supposed to be present and then the fish is searched by hand from an opening of the top of the trap. Fishes like *Clarias, Heteropneustes, Anabas* etc. are trapped commonly. Sometimes in deeper water two persons are needed for the operation of the traps.

Scooping. The rectangular, triangular or round bamboo frame is used and the net of same size and shape is loosely hung with this frame. The rounded net is generally used for transfer of fresh catch from the drag net.

Dip or lift net. The basic principle of lift net fishing is to keep the net in water for about 5 to 10 minutes, and then pull it out rapidly. In this method the fishes which are over the net at that time may be caught. Some baits are also put on the net or suspended over it. On this principle varying nets of different shapes and sizes are used in different regions. All these nets are fixed to frame for catching large and small sized fishes. Four types of net Helajal, Kharra Jal, Bhesal Jal and

Khoursla Jal are commonly used.

Cast net (Ghagaria jal). This is a circular umbrella-shaped net made up of cotton twine. The number of meshes on the mouth is 50 and at the periphery around 1,000. The size of the mesh is 2.5 cm and at the periphery circular pockets are formed by folding the net inwards to about 6.5 meshes in depth. Each pocket contains two iron sinkers. On its apex a thin rope of 3 mm diameter is woven through 50 meshes and it is tightened to close the mouth. The free end of the rope is tied to the fore finger of the left net. At the time of fishing (the net is swung overhead and dropped to the distance reachable by attached rope. Cast net is operated from the boat. After sometime it is slowly dragged in water with the rope. The fishes enter in the net pockets and thus are caught.

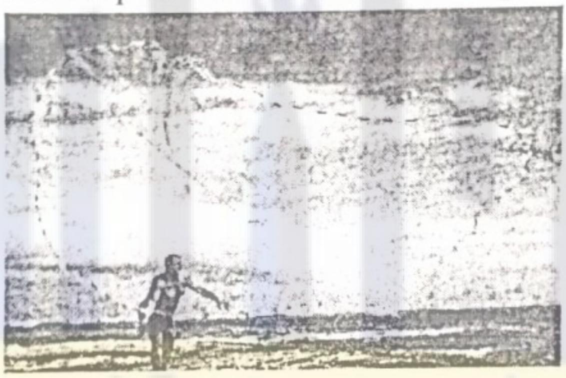

Fig 12.9: Caste Net

Purse net. Purse nets are generally used to capture the migratory fishes like **Hilsa** in the month of October to January and large sized carps and cal fishes are fished in May to July. In river Ganges purse net fishing is recorded. Two types of purse nets are operated in our country **i. e.**, Kharki Jal and Shangla Jal. The wide mouth has two flexible bamboo rods hinged at the two angles and forming upper and lower lips. The net is suspended in deep water and attached from the boat. At the time of operation the mouth is kept open but as fish enters into the net, mouth is closed due to pressure on the bamboo poles fig: purse net.

Kharki Jal (Purse Net)

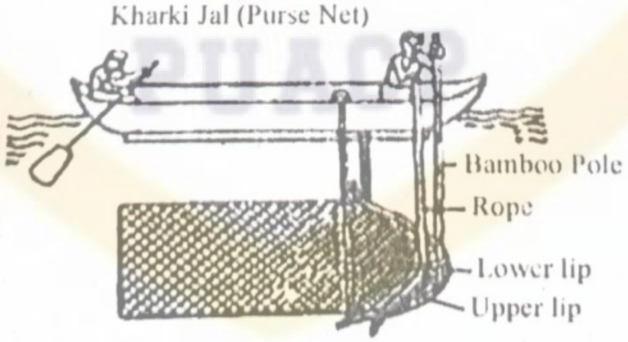

Fig 12.10: Purse net

Gill net. Gill nets are wall-like with a mesh opening of varying size with type of fish to be fished. The net is set on the transverse direction of the migrating fish. The net is made up of common hemp fibers but now-a-days nets are prepared from synthetic impermeable fibers due to which it is not possible for the fishes to see the nets. So, as fishes try to swim through a net wall the meshes from a noose round its head and they are caught. Three types of gill nets are commonly used viz., Set net (stationary nets) Floating gill net and Drift net. The net is set in the morning hours and caught fishes are collected in evening. The fishes caught by this net are Seenghala, Pangasius, Major carps, Silonia, etc. This net is fit for fishing in fast current.

Fig 12.11: Gill net

Drag net. This net is commonly used for fishing in lakes and rivers. It is provided with a main rope which carries floats and a foot rope bearing sinkers. The shape and size of the net is varying according to the water source where fishing is to be performed. Some drag nets are provided with peripheral pockets.

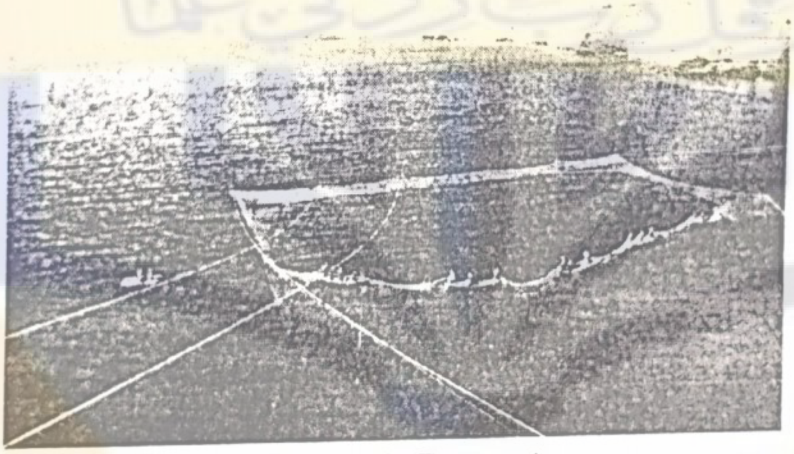

Fig 12.12: Drag net

Mainly two types of drag nets are constructed:

(1) The first one is consisting of a bag with two wings.

(2) The second is a wall like formed of a very long net of which the upper margin is supported by a strong rope and having a number of wooden floats. It extends from bank to bank and needs 25 to 30 fishermen to drag it.

It is hoped that expert and advanced technology would be able to design some net in future which may make fishing efficient and easy.

Electric Fishing

Although it is not of common use but sometimes it is used with precautions for fishing. It is reported that fishes are attracted towards electric field on electrode. The current is produced by generator or from batteries. The electric current is of about 15 amperes. The voltage may vary according to the depth, distance and conductivity of water. Large fishes are more sensitive than smaller ones. The voltage is between 150 to 250 volts. Direct current, alternating or interrupted current can be used. If the voltage and intensity is well regulated the fish soon recover from the shock in the electric field.

Preservation of Fish

If not preserved, fishes get spoiled due to decomposition. Fish flesh comprises of protein, fat, minerals, vitamins, amino acids, iodine, phosphorus and large quantity of water. Fishes are also provided with a number of bacteria which after the death, start to attack various constituents. In this light the preservation of fishes is an essential process. The following methods of preservation of fishes are generally practiced in India and abroad.

Refrigeration. The basic idea behind refrigeration is to preserve the fish at 0°C which prevents the spoilage for short period. For this purpose alternate layers of fish and ice are kept in closed vessels to maintain the temperature al 0°C. In case of large fishes ice pieces are kept in abdominal cavity of gutted fishes.

Deep-freezing. Before this freezing, fishes are washed properly and kept at a temperature of-18°C for longer period. Only the fresh fish in good condition are deep-frozen. Before keeping the fish in this process the heads of large fishes are removed. They are also gutted and washed. By this method fishes can be kept for longer period without spoilage.

Freeze drying. This is long process and expensive so only good quality of fishes is put to this type of preservation. The first step is the freezing of the fish, which is then allowed to dry by sublimation. In this process ice is changed into water vapor without melting. The color and nutritive substances are completely preserved by this technique. Further, the fish is frozen to -20°C by keeping them in freezing chamber. After freezing, fishes kept in trays are sent to the cabin containing horizontal heating plates for drying in vacuum. Now fishes are well dried due to hot plates, and packed in air-conditioned chamber.

Sun drying. Small sized fishes are dried in sun light especially in tropical countries like India, China, Japan and others, where sun rays are very powerful (scorching) to dry the fishes easily. The fishes are kept for dehydration on a mat or anything for 3 to 5 days and during this period turning over the fishes is continued. Larger **fishes** are cut into pieces for easy drying. But this method is not perfect for longer preservation.

Sun curing. An advanced method over simple sun drying is developed in which body of the fish is opened from the ventral side and the viscera and the gills are removed. Then the fish is washed and salted in ratio of 1: 3 to 1: 8 (salt: fish) which is related with the size of fish.

Wet curing. It is just like sun curing with the only difference in the packing of salted fish as such. This method is employed only for fatty fishes.

Salting. The partial dehydration of the fishes by osmosis with sodium chloride is termed as salting. If intense, this will kill the microbes and stop diastasis. The heads of the fishes are removed, gutted and washed then salted as soon as possible. Salting can be carried out as follows:

Dry (alternate layer of fish and salt)

In brine (light, 16% salt: strong, 25% salt).In brine after the fishes have been dipped in salt are cooled by spreading salt and crushed ice on the fish. For the light and cold salting, the processes, should be performed in cold room (2 to 3°C) but for strong salting this process can be performed at the normal room temperature.

Smoking. This preservation by the action of wood smoke is known as smoking of the fish. This permits the preparation of delicate specialties. The temperature of the smoke and its rate of circulation should be controlled. The smoking may be of hot or cold type. For cold smoking fishes are dried, salted, exposed to a smokeless fire (38°C) and then processed for real smoking at 28°C. The hot smoking is performed on fresh fishes in which fish is subjected at 130°C on strong fire which is followed by smoking at 40°C. The industrial circulation should be controlled. The smoking may be of hot or cold type. For cold smoking fishes are dried, salted, exposed to a smokeless fire (38°C) and then processed for real smoking at 28°C. The hot smoking is performed on fresh fishes in which fish is subjected at 130°C on strong fire which is followed by smoking at 40°C. The industrial smoking is done in galleries with a smoking installation and a system for the proper circulation of smokes. This method was used to preserve the fishes in World War II but it is not recommended in the present day fish industries.

Canning. The canning of fish is a long, complicated and advanced process in which head and viscera are first removed, eviscerated fishes are treated with brine, washed, dried and cooked in olive oil to remove the excess of water for 2 to 4 minutes only. Now cooked fishes are packed in olive oil in tins and 1 sealed .and then dispatched out for markets. This is very costly process so it is not in common use. This process of preservation is widely used in America.

◉ *Fish Culture and Water Pollution*

Water pollution is a serious event is many countries of the world where, along with the industrialization programmes, population pressure increase also takes place much faster than the treatment facilities. Effluents of fertilizer factories, pulp mills, textile, paper mills, distilleries, nuclear reactors, thermal plants and use of pesticides, herbicides, etc. change the quality of water j resources with regard to their chemical, physical and biological parameters. The fishes are directly influenced due to these toxic chemicals which cause mortality and several diseases as a result the growth is ceased. The breeding capacity of fishes is also hampered due to these toxicants. Pollution affects the fish culture programme directly and indirectly both as given:

(1) Change in physical, chemical and ecological nature of water resources.

(2) Decreased percentage of dissolved oxygen in water affects the respiratory function of gills.

(3) Cause reduction in phyto and zooplanktons.

(4) Destroy the spawning grounds and other nourishing materials.

(5) Fish flesh is loaded with pollutants.

(6) Flesh becomes unpalatable.

(7) Sewage and silt cause quick destruction of ponds.

Therefore, water resources should be kept free from these toxic pollutants which would be of great help for fish industry and human-beings also.

Composite Fish Farming

It is found that if few selected species of fishes are stocked together in proper proportion in a pond, total production of fishes is increased many times. This mixed fish farming is termed as composite fish farming or 'Polyculture'. The main idea behind this type of farming is given as:

(1) All available niches are fully utilized.

(2) Compatible species do not harm each other.

(3) No competition among different species is found.

(4) Fishes may have beneficial effect on each other.

Food material supplied in this type of farming is in accordance with the feeding habit of the different species. In India mixed fish culture is an old practice in which Catla carta, Labeo rohita and Cirrhina mrigala are surface feeder, column feeder and bottom feeder respectively and are used for composite fish farming.

Now-a-days in India polyculture is progressing well in larger ponds on commercial scale. In W. Bengal at Anjana fish Farm, 4,000 kg/hectare/year production of fish has been recorded. In Central Fisheries Research Institute the investigations are going on for best combination of fishes for polyculture in Indian water resources. It is hoped that successful steps of polyculture would j certainly increase the fish production rate in our country.

Exotic Fishes

The fishes imported into a country for fish culture are called as 'Exotic fishes' and such fish culture is known as 'Exotic fish culture'. A large number of exotic food fishes have been imported and a few of them have achieved good success for fish culture industry in India.

(1) The Gourami **(Osphronemus goratny)** is native of Indonesia and has been introduced in Calcutta. In 1809 same fish was introduced in the ponds off Chennai and Nilgiris from Mauritius. The Tench **(Tinca tinca)** was imported from England and introduced into the Ootacamund Lake. It is quite successful fish for fish culture.

(2) The Crucian carp **(Carassius carassius)** is a golden carp, native of central Europe. It was introduced in 1870 in Ooatacamund Lake and further transferred in ponds of Nilgiris.

(3) The common carp **(Cyprinus carpio)** is native of China and was introduced in Shri Lanka from Prussia in 1914.